Formalizing the interface between category learning and phonetics generates further predictions for explaining nasal vowel typology

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Introduction

Cross-linguistically, nasal contrasts occur more commonly on low vowels than high vowels, e.g. Amuzgo contrasts [a] and [ã], but has no nasal counterpart to [i] (Kingston 2007; Longacre 1966). Phonetic experiments on vowel height and nasality have found multiple parallel effects, with debate over which one(s) cause the typological asymmetry (Henderson 1984; Hajek 1997; Kunay et al. 2022). By computationally modeling the interface between phonetic effects and phonological category innovation, we identified a novel prediction for evaluating these candidate causes. As a case study, we test this prediction in a vowel discrimination experiment, providing evidence disfavoring one of the causes: the elevated perceptibility of nasality on long vowels.

Background

Low vowels tend to be longer than high vowels (Peterson and Lehiste 1960). Nasality is more perceptible on longer vowels (Whalen & Beddor 1989). Hajek (1997) proposes that this difference causes the typological asymmetry (henceforth the "length-perception hypothesis").

Contrastive vowel nasalization develops from nasal vowel allophony (*a → a and *an → ãn) and then consonant loss (*a → a and *ãn → ã) (Hajek 1997; cf. Beddor 2009). For nasal contrasts to develop more often in low (longer) than high (shorter) vowels, these changes would need to occur more often in low vowels. However, little is known about the exact sound change mechanisms and how this hypothesized phonetic cause would result in the required difference.

Model

We implemented a Dirichlet process Mixture of Gaussians model of learning and sound change (Kirby 2013; Gubian et al. 2023). The learner searches for the set of categories, and number of categories, that best fits its input data: uncategorized vowel tokens. A child's input consists of its parent's productions (noisy samples from the parent's categories) and it can learn categories different from its parent's. Nasal allophony arises when vowels that neighbor different consonants split into two categories, a nasal and an oral allophone. The less overlap between vowel distributions, the more likely the next generation is to learn them as separate categories. If a phonetic effect causes more nasal contrasts, then it should shift the oral consonant context and nasal consonant context distributions to overlap less in the learning input. Thus, the "length-perception hypothesis" predicts that there should be a bigger difference in the distributions of perceived nasality between nasal and oral consonant contexts, a criterion that has not been directly tested by previous phonetic experiments.

Experiment

We conducted a discrimination experiment manipulating the duration of vowel nasality, length, and consonant context (cf. Zellou 2017). Vowel stimuli with an ambiguous degree of nasality, likely falling in the region of overlap, allowed for studying perceptual distributions with only a limited set of stimuli. Mixed effects linear regression model estimation reveals that vowel length did not result in the ambiguous token being perceived as significantly more nasal preceding a nasal consonant, contra the "length-perception bias hypothesis."

Discussion

This finding directs future work toward other phonetic causes and empirically examining the model mechanisms (e.g. the precise role of distributional overlap, cf. Feldman 2013).